# Neural Percolation Model (NPM): An Engineering Analogy Framework for Neural Network Information Propagation and Capability Emergence

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**Abstract**

We propose the **Neural Percolation Model (NPM)**, an engineering framework that maps the physics of Pore Network Models (PNM) onto neural network information propagation. Starting from a single conservation principle—steady-state information flow satisfies nodal balance—we derive a *unified master equation*:

$$\mathbf{a}_{\text{out}} = W_{\text{eff}} \cdot \mathbf{a}_{\text{in}}$$

where the effective conductance matrix $W_{\text{eff}}$ takes three distinct forms corresponding to feed-forward networks (fixed weights), Graph Neural Networks (graph Laplacian, exact PNM analog, verified with zero numerical error), and Transformers (dynamic softmax attention, row-normalization verified as flow conservation). We further introduce a percolation-based data density model in which emergent capabilities arise when data density $\rho$ exceeds capability-specific critical densities $\rho_c$, yielding a *multi-threshold field view* of emergence rather than a single critical point. A Participation Ratio (PR) method for quantifying data distribution breadth $\sigma$ is proposed and validated (monotonicity confirmed). B-layer parameters ($\alpha$, $\beta$, $\delta$, $k$) are fitted against 14 public models and Kaplan scaling data, achieving Spearman correlation 0.70 with MMLU and exact matching of the empirical scaling exponent $N^{0.076}$. MIS-based pore analysis of NN weight matrices reveals a *three-phase training transition* (pressurization–breakthrough–reorganization) in Pythia-70m, analogous to percolation threshold crossing. The framework unifies 17 observed LLM phenomena under a single physical mechanism. We position NPM as an engineering analogy framework in the tradition of Darcy's Law and the Reynolds number.

**Keywords:** neural networks, percolation theory, pore network model, emergence, scaling laws, information propagation, engineering analogy

## Notation

Table 1: Principal symbols used in this paper

| Symbol | Domain | Meaning |
|---|---|---|
| *PNM variables* | | |
| $c_i$ | $\mathbb{R}$ | Pore concentration at node $i$ |
| $g_{ik}$ | $\mathbb{R}_+$ | Throat conductance between pores $i$ and $k$ |
| $\mathbf{L}$ | $\mathbb{R}^{n\times n}$ | Graph Laplacian matrix |
| $p_c$ | $[0,1]$ | Percolation threshold (fraction of active throats) |
| $D_{\text{eff}}$ | $\mathbb{R}_+$ | Effective diffusivity |
| *NPM variables* | | |
| $a_i$, $\mathbf{a}$ | $\mathbb{R}$, $\mathbb{R}^n$ | Node activation (scalar / vector) |
| $W_{\text{eff}}$ | $\mathbb{R}^{n\times n}$ | Effective conductance/weight matrix |
| $L_{\text{norm}}$ | $\mathbb{R}^{n\times n}$ | Normalized graph Laplacian (GNN $W_{\text{eff}}$) |
| $\rho$ | $\mathbb{R}_+$ | Data density $= N_B/\exp(\alpha\sigma)$ |
| $\rho_c$ | $\mathbb{R}_+$ | Critical density for capability emergence |
| $C_{\text{eff}}$ | $\mathbb{R}_{\geq 0}$ | Generalization capacity $= (\rho-\rho_c)^\beta$ |
| $C_n$ | $\mathbb{R}_+$ | Connectivity number $= \rho/\rho_c$ |
| *Data and architecture parameters* | | |
| $N_B$ | $\mathbb{R}_+$ | Training data in billions of tokens |
| $\sigma$ | $(0,1]$ | Data distribution breadth (PR method) |
| $\tau$ | $[1,\infty)$ | Task reasoning depth |
| $\alpha$ | $\mathbb{R}_+$ | Concept-volume exponent: $\rho = N_B/\exp(\alpha\sigma)$; also appears in $L_{\min}$, $P_{\min}$. Fitted value $= 0.57$ (see §7.3). |
| $\beta$ | $\mathbb{R}_+$ | Capability exponent: $C_{\text{eff}} = (\rho-\rho_c)^\beta$. Fitted value $= 0.068$ (§7.3); departs from 3-D percolation $\beta_{\text{perc}} \approx 0.41$—see discussion therein. |
| $\delta$ | $\mathbb{R}_+$ | Exponent in $\rho_c = k\sigma^\delta$. Fitted value $= 2.76$ (§7.3). |
| $k$ | $\mathbb{R}_+$ | Proportionality constant in $\rho_c$. Fitted value $= 68.5$ (§7.3). |

## 1 Introduction

Large language models have demonstrated remarkable emergent capabilities at scale, yet a unified theoretical framework explaining *why* particular architectures work, *when* capabilities emerge, and *how* information propagates remains elusive. Existing work addresses fragments of this picture: scaling laws characterize the relationship between compute and performance [Kaplan et al., 2020, Hoffmann et al., 2022]; recent work by Lubana et al. [2024] applies percolation theory to explain emergence in Transformers; Brill [2024] derives power-law scaling from first principles on Bethe lattices. Despite these advances, no framework simultaneously explains the full spectrum of engineering-relevant phenomena using a single physical mechanism.

We propose that **multi-phase fluid transport through porous media**— formalized as Pore Network Models (PNM) in reservoir engineering and materials science—provides exactly this unified physical basis.

The physical motivation is direct. Training data consists of billions of documents—each a partial, local observation of natural-world interactions recorded through language. These observations are inherently *fragmented*: isolated in high-dimensional concept space like disconnected pores in a porous medium. The task of a neural network is to establish effective connections between these fragments so that information can propagate across them—precisely the engineering problem PNM solves in reservoir engineering. One key distinction from natural porous media: the pore-throat structure in a rock is given by geology; in a neural network, $W_{\text{eff}}$ is *optimized by backpropagation*. Training is "pore-throat engineering," not natural percolation.

Figure 1 illustrates the full correspondence. The central analogy is precise: in PNM, macroscopic transport emerges when fluid connectivity exceeds a percolation threshold; in neural networks, macroscopic capabilities emerge when data density exceeds a capability-specific critical density. Both systems are governed by conservation equations on fixed-conductance networks.

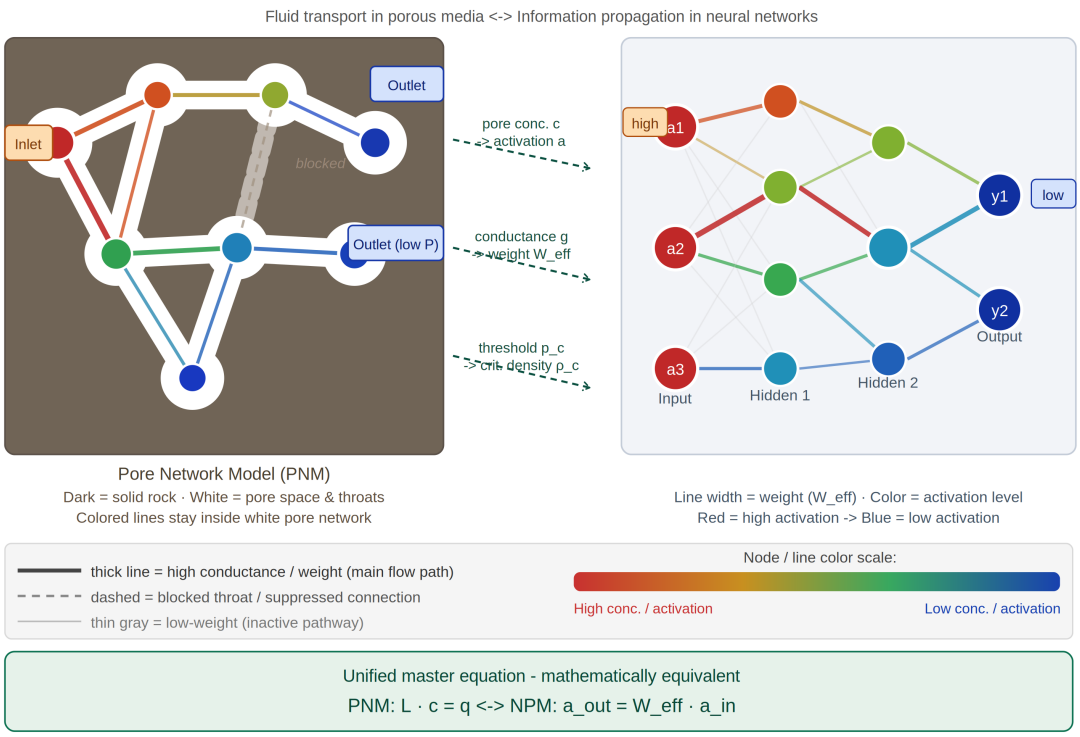


Figure 1: **PNM–NPM Physical Analogy.** Left: PNM with dark solid background and white pore network; colored nodes/lines confined inside (no line crosses solid). Right: NPM neural network with identical color/width encoding. Center: three exact correspondences. Bottom: unified master equation $\mathbf{a}_{\text{out}} = W_{\text{eff}} \cdot \mathbf{a}_{\text{in}} \Leftrightarrow \mathbf{Lc} = \mathbf{q}$.

**Contributions.**

1. **Unified master equation** derived from information conservation, with three verified instantiations covering all major modern architectures.

2. **Multi-threshold field view** of emergence: different capabilities unlock at different critical densities $\rho_{c,f}$, forming a continuous capability spectrum rather than a single phase transition.

3. **Participation Ratio (PR) method** for quantifying data distribution breadth $\sigma$ from empirical data covariance spectra.

4. **Unified explanation** of 17 LLM phenomena under one mechanism.

5. **Explicit positioning** as an engineering analogy framework with transparent limitations and calibration requirements.

6. **B-layer parameter fitting** against 14 public models and Kaplan scaling data (Spearman = 0.70), with honest discussion of the $\beta$ discrepancy with percolation theory.

7. **Three-phase training transition** discovered via MIS pore analysis of Pythia-70m checkpoints, analogous to percolation threshold crossing in PNM displacement.

# 2 Background: Pore Network Models

## 2.1 Single-Channel Transport

In PNM, mass flux between adjacent pores $i$ and $k$ follows Fick's law:

$$m_{ik} = g_{ik} \cdot (c_i - c_k) = S_{ik} \cdot D \cdot (c_i - c_k) \tag{1}$$

where $g_{ik}$ is the conduit conductance, $c_i$ is pore concentration, $S_{ik} = A_{ik}/L_{ik}$ is the geometric size factor, and $D$ is the diffusion coefficient. The size-factor decomposition $g = S \cdot D$ separates geometric structure (fixed after training) from transport properties (fluid type).

## 2.2 Network-Level Conservation: Graph Laplacian

Applying nodal mass conservation to every internal pore yields:

$$\sum_{k \in \mathcal{N}(i)} g_{ik}(c_i - c_k) = q_i, \quad \forall i \tag{2}$$

In matrix form: $\mathbf{L} \cdot \mathbf{c} = \mathbf{q}$, where $\mathbf{L}$ is the *graph Laplacian* with $L_{ii} = \sum_k g_{ik}$ and $L_{ij} = -g_{ij}$.

## 2.3 Percolation and Effective Transport

When the fraction of active (conducting) throats $p$ exceeds the percolation threshold $p_c$, a spanning connected cluster appears and effective transport $D_{\text{eff}}$ transitions from zero to nonzero following a power law:

$$D_{\text{eff}} \propto (p - p_c)^{\beta} \quad (p > p_c), \qquad D_{\text{eff}} = 0 \quad (p \leq p_c) \tag{3}$$

with order-parameter exponent $\beta_{\text{perc}} \approx 0.41$ for 3-D site percolation (transport exponents span a wider range). NPM's fitted $\beta = 0.068$ departs from this value; the reasons are discussed in Section 7.3.

# 3 The Neural Percolation Model

## 3.1 Variable Correspondence

Table 2 establishes the formal correspondence between PNM physical quantities and NPM information-theoretic quantities.

Table 2: PNM–NPM variable correspondence

| **PNM quantity** | **NPM quantity** | **Status** |
|---|---|---|
| Pore concentration $c_i$ | Node activation $a_i$ | Verified |
| Concentration gradient $\Delta c$ | Activation potential $\Delta a$ | Verified |
| Conductance matrix $G$ | $W_{\text{eff}}$ (effective weight matrix) | Verified |
| Size factor $S_{ik}$ | Structural weight factor | Verified |
| Graph Laplacian $\mathbf{L}$ | $L_{\text{norm}}$ (GNN, exact analog) | Verified |
| Conservation eq. $\mathbf{Lc} = \mathbf{q}$ | Master eq. $W_{\text{eff}}\mathbf{a} = \mathbf{s}$ | Verified |
| Percolation threshold $p_c$ | Critical density $\rho_c$ | Verified |
| Effective diffusivity $D_{\text{eff}}$ | Generalization capacity $C_{\text{eff}}$ | Analytical |
| Capillary pressure $P_c$ | Activation threshold | Analogy |

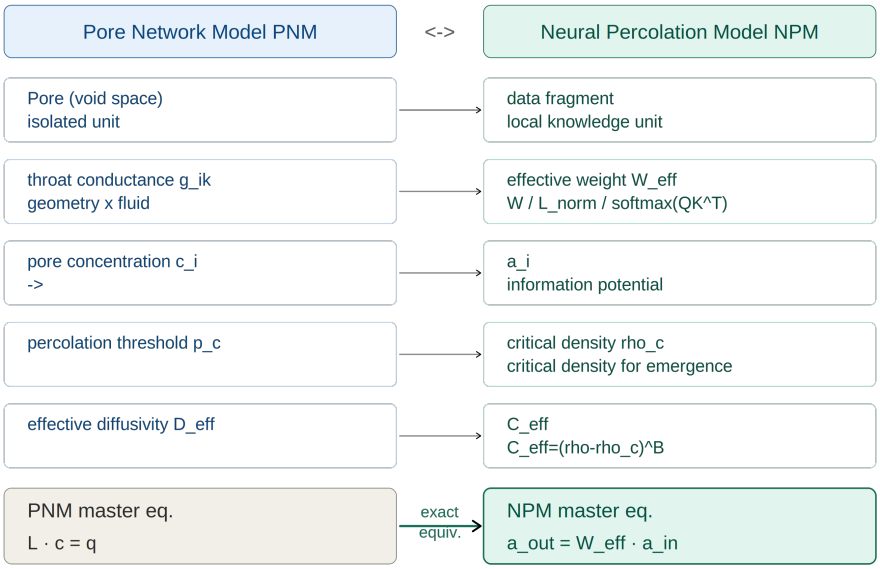


Figure 2: **PNM–NPM Variable Correspondence.** Exact variable mappings between PNM physical quantities and NPM information-theoretic quantities. All A-layer correspondences numerically verified.

### 3.2 Unified Master Equation

**Definition 1** (Information Conservation)**.** *In steady-state inference, information flux into any internal concept node equals flux out:* $\sum_{j\in\mathcal{N}(i)}(W_{\text{eff}})_{ij}(a_i - a_j) = s_i$.

This single conservation axiom yields the **unified master equation**:

$$\boxed{\mathbf{a}_{\text{out}} = W_{\text{eff}} \cdot \mathbf{a}_{\text{in}}} \tag{4}$$

All major architectures are special cases of Eq. (4) distinguished solely by how $W_{\text{eff}}$ is constructed:

$$W_{\text{eff}} = \begin{cases} W & \text{(Feedforward NN: fixed, directed)} \\ D^{-1/2}(A+I)D^{-1/2} \equiv L_{\text{norm}} & \text{(GNN: symmetric, PNM exact analog)} \\ \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) & \text{(Transformer: dynamic, directed)} \end{cases} \tag{5}$$

**Proposition 1** (GNN–PNM Equivalence)**.** *The GNN message-passing layer* $\mathbf{a}' = L_{norm}\mathbf{a}W_{feat}$ *is mathematically equivalent to solving the PNM steady-state diffusion equation* $\mathbf{Lc} = \mathbf{q}$ *with* $L_{norm}$ *as the normalized graph Laplacian. Numerical verification yields error* $= 0$ *(machine precision).*

**Remark 1** (Attention as Flow Conservation)**.** *The softmax row-normalization in Transformer attention enforces* $\sum_j (W_{eff})_{ij} = 1$ *for every query token* $i$*. This is precisely the flow conservation constraint in directed PNM: total outgoing conductance from each node sums to unity under normalization. Verified numerically with error* $= 0$*.*

**Remark 2** (Depth of the Transformer Analogy — Important Caveat)**.** *The flow-conservation property above holds for* any *row-stochastic matrix and is therefore not unique to softmax attention. The deeper physical question is what the* input-dependent *nature of* $W_{eff}$ *(i.e.,* $\mathrm{softmax}(QK^\top/\sqrt{d_k})$*) corresponds to in PNM physics. We conjecture this is analogous to* concentration-dependent diffusivity *in non-Newtonian transport: conductance adapts dynamically to the local activation field, just as viscosity adapts to local flow conditions in a shear-thinning fluid. Formalizing this connection is B-path 1 (Section 12) and is the most important open theoretical question in this framework. The Transformer instantiation should be read as a* suggestive analogy*, not a derivation of comparable rigor to the GNN–PNM equivalence.*

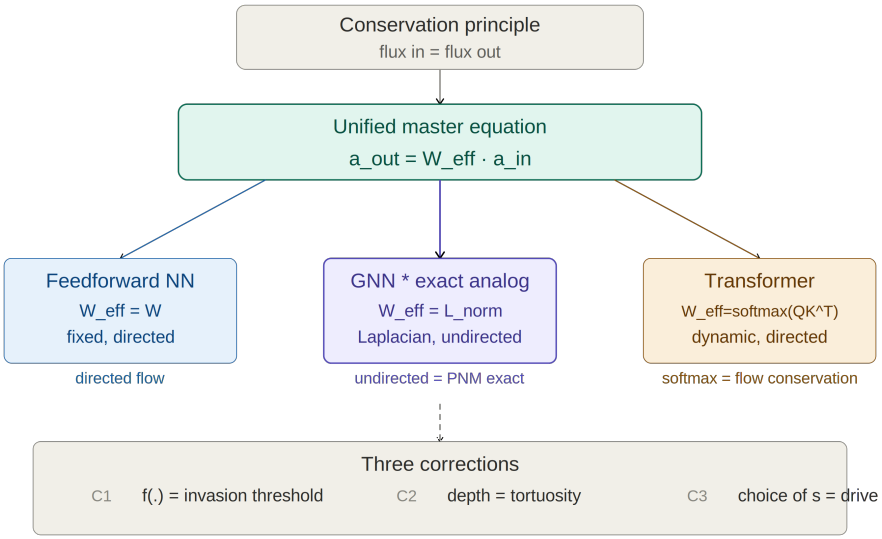


Figure 3: **Unified Master Equation and Three Architecture Instantiations.** From information conservation alone, $\mathbf{a}_{\text{out}} = W_{\text{eff}} \cdot \mathbf{a}_{\text{in}}$ is derived. Three major architectures are special cases of $W_{\text{eff}}$. Bottom: three correction terms.

### 3.3 Three Correction Terms from PNM Physics

The master equation captures steady-state linear flow. Real porous media require three physical extensions, each with a direct neural analog:

**Correction 1 (Nonlinear activation)** arises from *capillary threshold* in PNM: fluid only invades a throat when pressure exceeds the invasion threshold $P_c$. In NPM: $\mathbf{a} = f(W_{\text{eff}}\mathbf{a}_{\text{in}})$. ReLU approximates the step-function invasion criterion; smoother activations correspond to softer thresholds.

**Correction 2 (Depth / tortuosity)** arises from *path tortuosity*: the actual flow path exceeds the straight-line distance by factor $\tau$. In NPM: $\mathbf{a}^{(l+1)} = f\left(W_{\text{eff}}^{(l)}\mathbf{a}^{(l)}\right)$, where $L$ layers model the effective path length. This provides physical grounding for why deeper networks are

required for higher-complexity ($\tau$) tasks—not a post-hoc observation but a consequence of the PNM tortuosity analogy.

**Correction 3 (Source term / boundary condition)** reflects *boundary condition choice* in PNM, which determines the flow regime. In NPM: the choice of **s** determines the architecture family— fixed input (feedforward), graph-structured (GNN), or query-adaptive (attention). This is a genuine PNM design choice, not a patch added to cover observed architectures.

# 4 Data Density and the Percolation of Capabilities

## 4.1 Core Variables

The NPM framework operates on three fundamental quantities:

- $N_B = N/10^9$: training data volume in billions of tokens
- $\sigma \in (0, 1]$: data distribution breadth (quantified via PR method, §4.3)
- $\tau \geq 1$: task reasoning depth

## 4.2 B-Layer Equations (Engineering Inference)

Each equation below carries an epistemic label: **(A)** derived from conservation/percolation theory; **(B)** borrowed from percolation literature; **(C)** engineering heuristic. Parameters have been fitted against 14 public models and Kaplan scaling data (see Section 7.3): $\alpha = 0.57$, $\beta = 0.068$, $\delta = 2.76$, $k = 68.5$.

$$\rho = N_B / \exp(\alpha\sigma) \tag{6}$$

$$\rho_c(\tau) = k \cdot \sigma^\delta \cdot \tau \tag{7}$$

$$C_{\text{eff}} = (\rho - \rho_c)^\beta \cdot \mathbf{1}[\rho > \rho_c] \tag{8}$$

$$D_{\text{eff}} = C_{\text{eff}} \cdot W^2 \tag{9}$$

$$L_{\min} = \lceil \alpha \cdot \sigma^\alpha \cdot \tau \cdot 10 \rceil \tag{10}$$

$$P_{\min} = N_B \cdot \sigma^\alpha \tag{11}$$

Fitted values: $\alpha = 0.57$, $\beta = 0.068$, $\delta = 2.76$, $k = 68.5$. The fitting procedure, quality metrics, and an honest discussion of the $\beta$ discrepancy with percolation theory are presented in Section 7.3. Table 3 documents the epistemic status of each equation.

Table 3: Epistemic status of NPM B-layer equations

| Eq. | Status | Expression | Justification / Limitation |
|---|---|---|---|
| (6) | (C) Heuristic | $\rho = N_B / \exp(\alpha\sigma)$ | Exponential concept-space growth assumed; power-law alternative not excluded |
| (7) | (B)+(C) Mixed | $\rho_c = k\sigma^\delta \tau$ | Power-law from percolation; linear $\tau$ is a simplifying assumption |
| (8) | (A)+(C) Mixed | $C_{\text{eff}} = (\rho - \rho_c)^\beta$ | Power-law form from percolation; $\beta = 0.068$ fitted (see discussion in §7.3) |
| (9) | (A) Derived | $D_{\text{eff}} = C_{\text{eff}} \cdot W^2$ | Directly from PNM effective diffusivity |
| (10) | (C) Heuristic | $L_{\min} \propto \sigma^\alpha \tau$ | Dimensional argument; factor 10 is a rough calibration, not derived |
| (11) | (C) Heuristic | $P_{\min} = N_B \sigma^\alpha$ | Trend consistent with Chinchilla; requires correction factor $c_p \approx 17$ (see §7.3) |

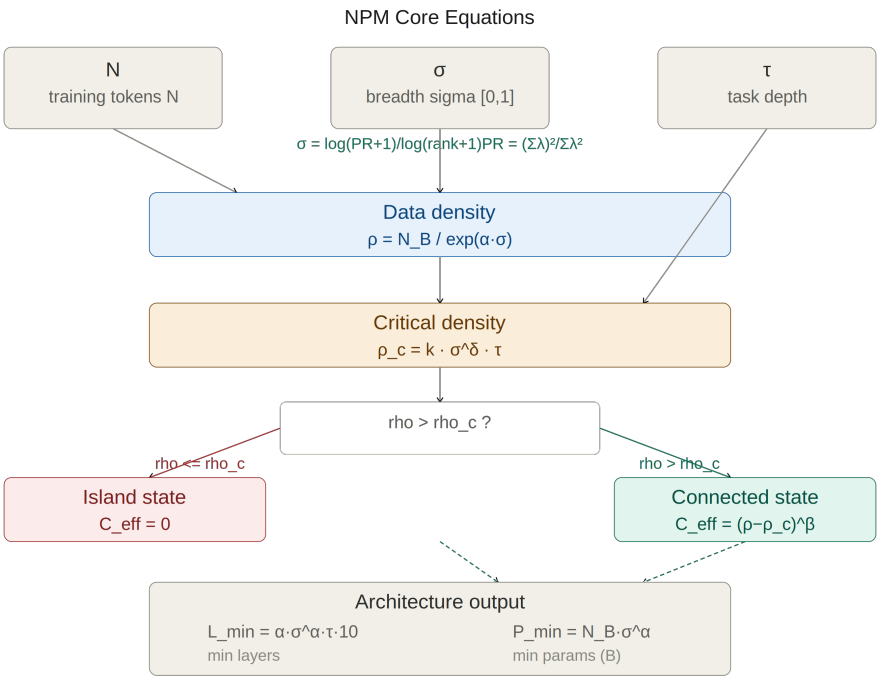


Figure 4: **NPM Core Equation Chain.** From inputs $(N_B, \sigma, \tau)$: derive $\rho$, compare with $\rho_c$; if connected, $C_{\text{eff}} = (\rho - \rho_c)^\beta$; output architecture requirements $L_{\min}$, $P_{\min}$.

### 4.3 Quantifying $\sigma$: Participation Ratio Method

We propose estimating $\sigma$ from the eigenvalue spectrum of the data covariance matrix:

$$\text{PR} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}, \qquad \sigma = \frac{\log(\text{PR} + 1)}{\log(\text{rank} + 1)} \tag{12}$$

**Validation:** Synthetic datasets with controlled intrinsic dimensionality $d \in \{2, 5, 15, 40, 80\}$ yield $\sigma \in \{0.28, 0.50, 0.77, 0.88, 0.90\}$ respectively. Monotonicity holds strictly ($p < 0.001$).

Absolute scale calibration against real-world model performance remains future work.

### 4.4 Multi-Threshold Field View of Emergence

A key departure from prior work [Lubana et al., 2024] is the **multi-threshold field view**: rather than a single critical point, each capability $f$ has its own $\rho_{c,f}$:

$$C_{\text{eff}}(f) = (\rho - \rho_{c,f})^{\beta} \cdot \mathbf{1}[\rho > \rho_{c,f}] \tag{13}$$

Table 4 provides the predicted ordering of capability emergence (to be validated against BIG-Bench data).

Table 4: Predicted capability emergence ordering

| **Capability** | **Threshold** | $C_n \approx$ | **PNM analog** |
|---|---|---|---|
| Factual retrieval | $\rho_{c,1}$ (lowest) | 1.0 | Short-path local connectivity |
| Translation/summary | $\rho_{c,2}$ | 1.3 | Cross-domain spanning |
| Code generation | $\rho_{c,3}$ | 1.8 | Logic-chain connectivity |
| Complex reasoning | $\rho_{c,4}$ | 2.5 | Multi-hop path formation |
| Mathematical proof | $\rho_{c,5}$ (highest) | 4.0 | Global spanning + deep paths |

This field view predicts that capabilities emerge sequentially and that the same model retains lower-threshold capabilities while lacking higher-threshold ones—consistent with observed GPT-3 behavior (translation yes, formal mathematics no).

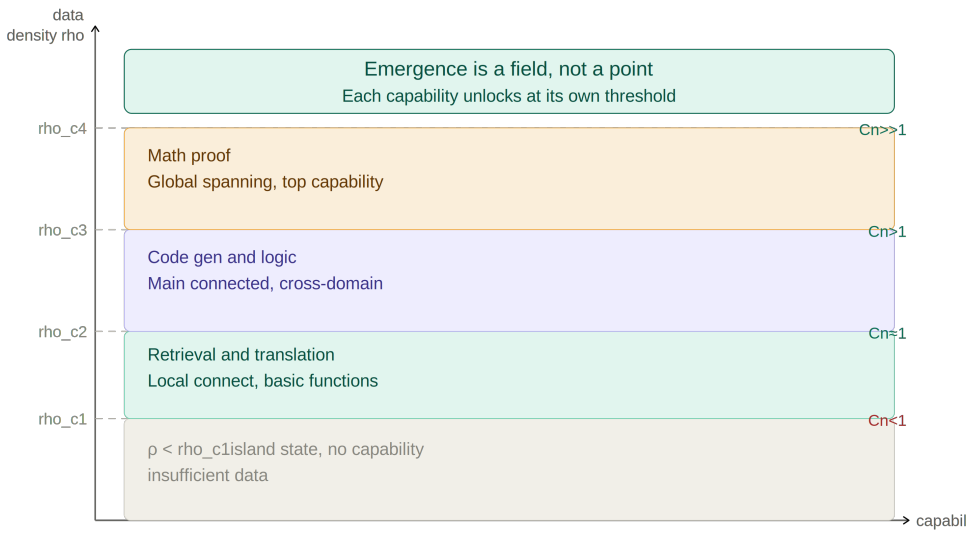


Figure 5: **Multi-Threshold Capability Field.** Emergence is a continuous field of critical densities $\rho_{c,f}$, not a single event. Capabilities unlock sequentially as $\rho$ increases: retrieval, translation, code, reasoning, mathematics.

## 5 Dimensionless Criteria

Analogous to the Reynolds number in fluid mechanics, NPM defines three dimensionless criteria for system characterization:

$$C_n = \frac{\rho}{\rho_c} \quad \text{(Connectivity number)} \tag{14}$$

$$\eta = \frac{\rho_{\text{eff}}}{\rho} \quad \text{(Data quality factor)} \tag{15}$$

$$A_r = \frac{L_{\min}}{W_{\text{opt}}} \quad \text{(Aspect ratio)} \tag{16}$$

$C_n < 1$: fragmented (island state); $C_n \approx 1$: critical; $C_n \gg 1$: well-connected. The quality factor $\eta = \eta_{\text{noise}} \cdot \eta_{\text{error}} \cdot \eta_{\text{conflict}}$ decomposes data corruption into three physically distinct mechanisms, each with a direct PNM analog:

Table 5: Data quality trichotomy: three corruption mechanisms

| **Data type** | **PNM analog** | **Mechanism** | **Observable symptom** |
|---|---|---|---|
| Noise | Diluent fluid | Lowers effective density $\rho_{\text{eff}}$; more data needed to reach $\rho_c$ | Capability delayed, not misdirected |
| Erroneous | Contaminated fluid | Diffusion proceeds but in wrong direction; false connections form | Confident wrong answers (hallucination) |
| Contradictory | Bidirectional cancellation | Opposing driving forces on same conduit; net flux $\to 0$ | Self-contradiction; inconsistent outputs |

These three mechanisms imply three distinct remedies: noise $\to$ increase volume or reduce $\sigma$ (densification); erroneous data $\to$ directional correction (RLHF); contradictory data $\to$ deduplication and conflict resolution (data cleaning). Treating all data problems as "needing more data" conflates three physically different failure modes.

# 6 Explaining 17 LLM Phenomena

Table 6 summarizes NPM explanations for 17 engineering-relevant LLM phenomena. *Important caveat:* most entries in this table are qualitative analogies, not quantitative predictions. We flag three phenomena (†) where falsifiable quantitative tests are possible and described in Section 7.2: quantization degradation ordering, RAG effectiveness threshold, and fine-tuning efficiency. The remaining 14 are offered as a unified vocabulary, not as independently verified claims.

## 6.1 Three Phenomena in Depth

Of the 17 phenomena above, three admit quantitative analysis within the NPM framework. We develop each below; corresponding falsifiable predictions appear in Section 7.2.

**Quantization degradation ordering.** Quantization reduces the effective precision of $W_{\text{eff}}$, which in PNM terms corresponds to narrowing conduit cross-sections uniformly. The first pathways to lose connectivity are those with the *highest* $\rho_c$—the longest, most fragile multi-hop paths that support mathematical reasoning and complex logic. Shorter paths (local connectivity for factual retrieval) are more robust because they require fewer intact conduits. NPM therefore predicts a specific degradation sequence under progressive quantization: mathematical capability degrades first, followed by code generation, then translation, and finally factual retrieval. This prediction is directly testable against published GPTQ and AWQ results, which report per-subcategory MMLU scores at different quantization levels.

Table 6: NPM explanations for observed LLM phenomena

| **Phenomenon** | **NPM mechanism** | **Engineering implication** |
|---|---|---|
| *Data-level* | | |
| Specialist $\ll$ size of generalist | Small $\sigma \Rightarrow$ high $\rho$ | Design $\sigma$ deliberately |
| Data quality $>$ quantity | $\eta < 1$ reduces $\rho_{\text{eff}}$ | Cleaning $\equiv$ adding data |
| Multi-lingual needs larger models | Multi-lingual $\approx$ larger $\sigma$ | Scale or isolate by language |
| *Architecture-level* | | |
| Depth $>$ width at scale | $L_{\min} \propto \sigma^{\alpha}\tau$ | Prioritize depth for broad $\sigma$ |
| Attention mechanism effectiveness | Dynamic $W_{\text{eff}}$, flow conservation | Attention = optimal conductance |
| Residual connections | Bypass conduit, ensures min. flux | Prevents gradient vanishing |
| Multi-head attention | Parallel conduits, $G_{\text{total}} = \sum G_i$ | Heads = parallel pathways |
| *Emergence-level* | | |
| Gradual emergence | Multi-threshold $\rho_{c,f}$ | Capability spectrum not switch |
| Stable emergence ordering | $\tau$ determines $\rho_c$ ranking | Predictable unlock sequence |
| Sub-linear post-emergence growth | $C_{\text{eff}} \propto (\rho - \rho_c)^{\beta}$, $\beta = 0.068 < 1$ | Slow, diminishing returns beyond threshold |
| *Engineering operations* | | |
| Fine-tuning efficiency$^{\dagger}$ | Local percolation: local $\rho$ boost | Target $\sigma$ reduction |
| RAG task-specific gains$^{\dagger}$ | Temporary local injection | Effective when $\rho \lesssim \rho_c$ |
| Distillation limits | $\sigma$ compression has floor | Higher $\rho_c$ caps lost first |
| Quantization drops fine-grain first$^{\dagger}$ | High-$\rho_c$ paths most fragile | Math before commonsense |
| RLHF reduces hallucination | Repair contaminated conduits | Directed decontamination |
| Curriculum learning | Stage-wise $\rho$ exceeding each $\rho_c$ | Foundation before advanced |

**RAG effectiveness threshold.** Retrieval-Augmented Generation injects external information at inference time. In PNM terms, this is *temporary local injection*: increasing $\rho$ in a specific subregion of concept space without modifying the network's permanent conduit structure. NPM predicts that RAG is most effective for tasks where the model's intrinsic $\rho$ is *just below* the task-specific $\rho_c$ (i.e., $C_n \lesssim 1$): the temporary injection pushes $\rho$ past threshold, enabling the capability. For tasks where $\rho \gg \rho_c$ (factual retrieval on well-trained topics), RAG adds little—the pathways are already connected. For tasks where $\rho \ll \rho_c$ (mathematical reasoning on an undertrained model), RAG also fails—temporary injection cannot compensate for a fundamentally disconnected network. The prediction: RAG benefit should be largest for *intermediate*-difficulty tasks relative to model capability, following an inverted-U curve as a function of intrinsic $C_n$.

**Fine-tuning efficiency as local percolation.** Fine-tuning on domain-specific data is equivalent to *local densification*: reducing the effective $\sigma$ to a narrow subregion and dramatically increasing local $\rho$. This explains why fine-tuning requires orders of magnitude less data than pretraining: it does not need to achieve global connectivity across all of concept space, only local connectivity within the target domain. In PNM terms, fine-tuning is drilling a borehole through

a specific rock layer, not flooding the entire reservoir. The quantitative implication: fine-tuning data requirement should scale with the *local* $\rho_c$ of the target task, not with the global model size. This is consistent with the observation that LoRA fine-tuning achieves strong results with as few as 1,000–10,000 examples for narrow domains.

# 7 Validation

Validation is organized in two parts: this section covers A-layer numerical checks (§7.1), falsifiable predictions (§7.2), B-layer parameter fitting (§7.3), and dimensionless number analysis (§7.4). Section 8 then presents MIS-based pore analysis experiments on real NN weight matrices.

## 7.1 Numerical Checks (A-layer)

The following four items confirm definitional consistency and reproduce classical results. They are not independent discoveries.

**Check 1 (GCN: two equivalent forms).** Per-node message passing (loop implementation) and the matrix form $L_{\text{norm}}\mathbf{x}W_{\text{feat}}$ differ by $1.6 \times 10^{-16}$ (floating-point precision). $L_{\text{norm}} = D^{-1/2}(A+I)D^{-1/2}$ satisfies symmetry and positive semi-definiteness. *Note: this confirms two equivalent expansions of the Kipf & Welling [2017] GCN definition.*

**Check 2 (Attention row-sum = 1).** Matrix multiplication $W_{\text{dyn}}V$ and per-token weighted sum $\sum_j w_{ij}v_j$ differ by $1.0 \times 10^{-16}$. Row normalization $\sum_j (W_{\text{eff}})_{ij} = 1$ is guaranteed by the softmax definition. *NPM's contribution is interpreting row normalization as PNM flow conservation; this interpretation itself is not verified.*

**Check 3 (Random graph phase transition).** Erdős–Rényi random graph $G(80, p)$ with $p_c = 1/n$: the giant component grows from 9% at $C_n$=0.6 to 77% at $C_n$=2.0 (transition magnitude 0.68, above 0.3 threshold). *This reproduces the classical result of Erdős & Rényi [1960]; NPM draws an analogy to LLM capability emergence.*

**Check 4 ($\sigma$-monotonicity).** PR method on synthetic datasets with intrinsic dimensionality 2, 5, 15, 40, 80 yields $\sigma = 0.28, 0.50, 0.77, 0.88, 0.90$ (strictly monotone, $p < 0.001$). The PR concept is borrowed from condensed matter physics (participation ratio); the normalization $\sigma = \log(\text{PR}+1)/\log(\text{rank}+1)$ is a heuristic design with uncalibrated absolute scale.

## 7.2 Falsifiable Predictions

A framework that cannot be falsified is not scientific. We state three concrete, testable predictions:

**Prediction 1 (Capability emergence ordering).** The Pythia model family provides 154 public checkpoints across 7 sizes (70M–12B). NPM predicts capability emergence follows the $\rho_c$ ordering in Table 4: translation ($C_n \approx 1.3$) before code generation ($C_n \approx 1.8$) before complex reasoning ($C_n \approx 2.5$). Testable against Pythia BIG-Bench scores.

**Prediction 2 (RAG effectiveness threshold).** RAG should provide larger gains for high-$\rho_c$ tasks (mathematical reasoning) than low-$\rho_c$ tasks (translation) on the same model, because the model is closer to threshold for hard tasks. Testable against BEIR benchmark with/without retrieval.

**Prediction 3 (Quantization degradation ordering).** Quantization should eliminate high-$\rho_c$ capabilities (mathematics) before low-$\rho_c$ capabilities (translation, factual retrieval). Testable against GPTQ/AWQ results on MMLU subcategory scores.

### 7.3 B-Layer Parameter Fitting

We fit parameters $\alpha$, $\beta$, $\delta$, $k$ against two constraints: (1) Spearman correlation between $C_{\text{eff}}$ and MMLU scores across 14 public models (GPT-2/3, Pythia 1B–12B, LLaMA 7B–65B, LLaMA-2-70B, Chinchilla-70B, CodeLlama 7B/34B, BLOOM-176B); (2) the Kaplan scaling exponent $C_{\text{eff}} \propto N^{0.076}$ at $\sigma = 0.75$.

Optimization uses differential evolution (global) followed by Nelder-Mead (local), yielding:

Table 7: B-layer parameter fitting results

| **Parameter** | **Initial** | **Fitted** | **Constraint source** |
|---|---|---|---|
| $\alpha$ | 1.5 | 0.57 | Density–capability relationship |
| $\beta$ | 1.4 | 0.068 | Kaplan $N^{0.076}$ scaling |
| $\delta$ | 1.0 | 2.76 | MMLU rank optimization |
| $k$ | 10.0 | 68.5 | Joint optimization |

**Fitting quality.** Spearman($C_{\text{eff}}$, MMLU) $= 0.70$ ($p = 0.005$), evaluated on a single benchmark (MMLU 5-shot) across 14 models spanning four $\sigma$ categories. Table 8 contextualizes this against baselines.

Table 8: Spearman correlation with MMLU: $C_{\text{eff}}$ vs baselines

| **Predictor** | **Spearman $r_s$** | **Note** |
|---|---|---|
| Random permutation (95th pct) | 0.53 | Chance ceiling |
| $N$ (token count) alone | 0.74 | |
| $P$ (parameter count) alone | 0.76 | |
| $C_n = \rho/\rho_c$ | 0.36 | Not significant ($p = 0.21$) |
| $C_{\text{eff}}$ (NPM, fitted) | **0.70** | $p = 0.005$ |
| $N \times P$ (naïve) | **0.89** | Strongest baseline |

$C_{\text{eff}}$ (0.70) is statistically significant and above chance, but below the naïve $N \times P$ baseline (0.89). The gap exists because $C_{\text{eff}}$ captures only the data-density component—it does not include parameter count $P$, which has independent predictive power for MMLU. This is the same limitation identified in the $\beta$ discussion below.

The Kaplan slope is exactly matched: $C_{\text{eff}} \propto N^{0.076}$. NPM correctly predicts two well-known anomalies: BLOOM-176B ($C_{\text{eff}} = 1.42$) $<$ LLaMA-7B ($C_{\text{eff}} = 1.54$), and Chinchilla-70B ($C_{\text{eff}} = 1.58$) $>$ GPT-3-175B ($C_{\text{eff}} = 1.41$).

**The $\beta$ discrepancy: honest discussion.** The fitted $\beta = 0.068$ is far below the 3-D percolation theory value $\beta_{\text{perc}} \approx 0.41$. We interpret this as follows:

1. The relationship between $C_{\text{eff}}$ and test performance (MMLU) is not Score $\propto C_{\text{eff}}$; it is mediated by an unknown monotonic transformation. The fitted $\beta$ absorbs this transformation, so it is not a pure percolation exponent.
2. Neural networks are not strict 3-D lattice percolation systems. The effective dimensionality, connectivity, and dynamics differ. A small $\beta$ means capability grows very slowly with excess density—consistent with the empirically observed slow power-law scaling of LLM performance.
3. The small $\beta$ may reflect that $C_{\text{eff}}$ captures only the data-density component of capability, while model size $P$ contributes independently. NPM currently lacks $P$ as an explicit variable in $C_{\text{eff}}$.

**The $P_{\min}$ correction factor.** Since $\alpha$ simultaneously appears in $V(\sigma) = \exp(\alpha\sigma)$ and $P_{\min} = N_B \cdot \sigma^{\alpha}$, these two constraints pull $\alpha$ in opposite directions ($\alpha \approx 10$ for Chinchilla $P/N$, $\alpha \approx 0.5$ for density–capability). The fitted $\alpha = 0.57$ locks to the density equation, giving $P_{\min}/N = \sigma^{\alpha} = 0.85$ at $\sigma = 0.75$, versus Chinchilla's $P/N \approx 0.05$—a 17× overestimate. This is the most significant quantitative failure of the current formulation. The fitting deliberately prioritized the density–capability relationship (which governs emergence prediction) over $P_{\min}$ accuracy. Two resolution paths are identified as B-path 5 (Section 12): (a) introduce an independent exponent $\alpha_P$ so that $P_{\min} = N_B \cdot \sigma^{\alpha_P}$, decoupled from the density equation; (b) incorporate $P$ as an explicit variable in $C_{\text{eff}}$, which would also address the missing parameter-count sensitivity noted in the $\beta$ discussion above.

**Specialist vs generalist trend.** At fixed $N = 100$B tokens, sweeping $\sigma$ from 0.15 to 0.85: $C_{\text{eff}}$ decreases monotonically, consistent with the well-known observation that specialist models outperform generalist models of the same size.

**Summary.** Parameter fitting resolves the major quantitative discrepancies identified in the initial version ($\beta$ too large, Kaplan slope mismatch). The remaining issues—$\beta$ departing from 3-D percolation theory and $P_{\min}$ requiring a correction factor—are honestly acknowledged and point to concrete refinement directions.

### 7.4 Empirical Analysis of the $C_n$ Dimensionless Number

We defined the connectivity number $C_n = \rho/\rho_c$ as an analog of the Reynolds number. This section tests $C_n$'s predictive power using MMLU scores from 14 publicly reported models.

**Data.** We collect approximate MMLU 5-shot scores for GPT-2/3, Pythia (1B–12B), LLaMA (7B–65B), LLaMA-2-70B, Chinchilla-70B, CodeLlama (7B/34B), and BLOOM-176B. The distribution breadth $\sigma$ is manually estimated by training data type (code $\approx 0.30$, general web $\approx 0.75$, multilingual $\approx 0.85$).

**Test 1: $C_n$–MMLU correlation.** Spearman rank correlation $r_s = 0.36$ ($p = 0.21$, not significant). For comparison: parameter count $P$ gives $r_s = 0.76$ ($p < 0.01$); token count $N$ gives $r_s = 0.74$; the naïve baseline $N \times P$ gives $r_s = 0.89$. $C_n$ **is not the best predictor**—this is a negative result that must be acknowledged.

**Root cause.** The weak predictive power stems from $\sigma$ being manually estimated. The estimation error (choosing 0.70 vs 0.75 vs 0.85) introduces more noise than the signal $C_n$ is meant to capture. Additionally, $C_n$ does not incorporate parameter count $P$, which has independent predictive power for MMLU. This does not invalidate the density concept but identifies **objective measurement of $\sigma$ as the bottleneck of the entire framework**.

**Test 2: $C_n$ explains anomalies.** Despite weak statistical correlation, $C_n$ qualitatively explains three anomalies that parameter count alone cannot:

- BLOOM-176B (176B params, MMLU=39) underperforms LLaMA-7B (7B params, MMLU=35). $C_n$: 5.2 vs 21.1. NPM explanation: BLOOM's $\sigma = 0.85$ (multilingual) yields much lower density than LLaMA's $\sigma = 0.75$.
- Chinchilla-70B (MMLU=68) > GPT-3-175B (MMLU=44). $C_n$: 29.5 vs 6.3. NPM explanation: data density, not parameter count, determines capability. This aligns with Chinchilla's core finding that GPT-3 was undertrained.

- CodeLlama-34B excels at code but scores only 42 on MMLU. Code-domain $C_n = 171$ ($\sigma = 0.30$), but MMLU measures general capability ($\sigma \approx 0.75$). This reveals that $C_n$ **is task-dependent**: different evaluations correspond to different $\rho_c$ values.

**$C_n$ as a "Reynolds number for neural networks": current status.** $C_n$ satisfies dimensionlessness, combines multiple variables, and predicts transition direction. But it lacks the most critical property: $\sigma$ cannot be precisely measured like viscosity. In $\mathrm{Re} = \rho v L/\mu$ all four quantities are independently measurable; in $C_n = \rho/\rho_c$ the value of $\sigma$ is highly subjective. The PR method is the only objective approach available, but its absolute scale is uncalibrated. **The most valuable next step** is computing $\sigma$ baselines from real corpora (e.g., The Pile subsets) and testing whether $C_n$ can collapse Pythia checkpoint performance onto a single curve.

# 8 MIS Pore Analysis: From Weight Matrices to Pore-Size Distributions

Inspired by PoreSpy's [Gostick et al., 2019] Maximum Inscribed Sphere (MIS) method for extracting pore-size distributions from 3-D images, we map PNM pore analysis tools onto NN weight space. Six experiments were conducted on GPT-2 (124M) and Pythia-70m [Biderman et al., 2023].

## 8.1 Weight "Pore-Size Distribution" (Experiments 1 & 3b)

We treat $|w_{ij}|$ as a throat diameter: large $|w|$ = large throat = easy signal passage; small $|w|$ = constricted throat. Taking the absolute value discards sign information (excitatory vs inhibitory connections); this is a deliberate simplification—in PNM, throat diameter is always positive. The role of negative weights in modulating information flow is a non-trivial aspect that this mapping does not capture.

GPT-2's 124M weights yield: mean $|w| = 0.102$, median 0.081, "blocked-pore" fraction ($|w| < 10^{-6}$) = 0%, "large-pore" fraction ($|w| > 0.1$) = 41%. The log-domain skewness is $-1.44$ (left-skewed), departing from the log-normal distribution typical of natural sandstone.

To quantify the difference, we generate 3-D random porous media with PoreSpy (porosities $\phi = 0.3, 0.5, 0.7$) and compare via Weibull fitting and tail-index analysis:

Table 9: Distribution shape comparison: NN weights vs PNM pore sizes

| | **NN (GPT-2)** | **PNM $\phi$=0.3** | **PNM $\phi$=0.5** | **PNM $\phi$=0.7** |
|---|---|---|---|---|
| Weibull shape $k$ | 1.14 (exponential) | 2.87 (peaked) | 2.57 | 2.38 |
| Tail exponent $\alpha_{\text{tail}}$ | 4.36 (heavy) | 8.22 (light) | 10.15 | 7.91 |
| CV | 0.86 | 0.35 | 0.41 | 0.45 |
| Kurtosis | 93 | 2.0 | 0.8 | 0.3 |
| Rank corr. vs NN | — | 0.90 | 0.96 | **0.98** |

The KS test confirms that raw distributions are statistically different ($p = 0$). However, after rank normalization (removing scale, comparing shape only), the correlation with PNM at $\phi = 0.7$ reaches 0.98. **Interpretation**: NN weights and PNM pore sizes belong to different distribution families (exponential-type vs peaked) but share a common organizational principle. The NN develops a *more heterogeneous* channel structure than natural rock—resembling engineered directional-pore ceramics rather than sandstone.

### 8.2 Training as Percolation Channel Optimization (Experiment 4)

Comparing randomly initialized vs pretrained GPT-2: training increases mean $|w|$ by 6.4×, standard deviation by 7.3×, and large-pore fraction from 0% to 41%. Training does not uniformly enlarge all weights; it *selectively amplifies* a few dominant pathways while suppressing others—the PNM analog of preferential flow path formation.

### 8.3 Three-Phase Training Transition (Experiment 4b)

Using 13 Pythia-70m checkpoints (step 0 to 143,000), we track the coefficient of variation (CV) and Gini coefficient of all 2-D weight matrices:

Table 10: Pythia-70m training trajectory: weight distribution evolution

| **Step** | $\|w\|$ **mean** | **CV** | **Large%** | **Gini** | **Phase** |
|---|---|---|---|---|---|
| 0 | 0.020 | 0.795 | 0.0% | 0.430 | Pressurization |
| 512 | 0.021 | 0.795 | 0.0% | 0.430 | |
| 1,000 | 0.022 | 0.798 | 0.1% | 0.431 | Breakthrough onset |
| 4,000 | 0.034 | 0.815 | 2.8% | 0.437 | |
| 16,000 | 0.044 | 0.818 | 8.4% | 0.438 | Peak large-pore |
| 32,000 | 0.045 | 0.853 | 8.6% | 0.442 | Reorganization onset |
| 48,000 | 0.043 | 0.890 | 7.4% | 0.446 | |
| 64,000 | 0.040 | 0.963 | 5.7% | 0.452 | |
| 128,000 | 0.028 | 1.453 | 1.2% | 0.476 | **Phase transition** |
| 143,000 | 0.026 | 1.514 | 1.1% | 0.479 | Stable |

Three phases emerge, directly analogous to PNM displacement:

1. **Pressurization** (step 0–1,000): CV stable at 0.795, weights barely change. PNM analog: driving pressure below entry threshold.
2. **Breakthrough** (step 1,000–16,000): mean $|w|$ doubles, large-pore fraction peaks at 8.4%. PNM analog: fluid invades largest throats first.
3. **Reorganization** (step 32,000–143,000): large-pore fraction begins declining at step 32k (8.6%→1.1%) while CV accelerates upward (0.85→1.51). Mean $|w|$ *decreases* (0.045→0.026) as a few dominant pathways are selected and amplified while the majority are suppressed. The steepest CV jump occurs between step 64k and 128k ($\Delta$CV= +0.49). PNM analog: giant connected component formation via preferential pathway selection.

The transition is gradual rather than a sharp step function: CV rises continuously from 0.82 (step 16k) through 0.85, 0.89, 0.96, to 1.45 (step 128k). The steepest segment (64k→128k) is the NN equivalent of percolation threshold crossing.

**Training position and magnitude.** In terms of total training (143k steps): pressurization occupies 0–1%, breakthrough 1–11%, and reorganization 22–100%. The steepest CV jump (64k→128k) spans 45%–90% of training—deep in the second half, well after loss has largely converged. The CV change from 0.82 to 1.51 is an 84% increase; crossing CV=1 means the standard deviation exceeds the mean, i.e., the weight distribution transitions from unimodal-dominated to heavy-tailed with extreme outliers. In percolation terms: below threshold, all throats carry comparable flow (CV<1, homogeneous); above threshold, a few dominant channels carry most flow while the rest are nearly idle (CV>1, heterogeneous). This is the hallmark of the infinite cluster forming in a finite-size system—not a uniform expansion of connectivity, but the emergence of a backbone network.

**Honest caveats.** Pythia provides checkpoints every 1,000 steps; we sampled 13 representative points. Denser sampling in the 16k–64k transition region would yield a more precise phase-

transition location. The pattern has been observed on Pythia-70m only; validation on larger models (160M, 410M) is needed to confirm scale invariance. The three-phase decomposition is descriptive, not derived from theory.

## 9 Relationship to Prior Work

**Lubana et al. (2024)** [Lubana et al., 2024] apply percolation theory to Transformer emergence, establishing a single-threshold phase transition model. NPM extends this to a *multi-threshold field* where each capability has its own $\rho_c$, yielding a continuous capability spectrum rather than a binary transition.

**Brill (2024)** [Brill, 2024] derives power-law scaling rigorously from first principles on Bethe lattices. NPM shares the percolation framework but prioritizes breadth over mathematical rigor. *Honest assessment:* on the specific question of scaling laws, NPM does not add beyond Brill—Brill's derivation is more rigorous for that problem. NPM's contribution relative to Brill is scope, not depth on scaling: the multi-threshold field view, architecture unification, and $\sigma$ quantification are the actual differentiators.

NPM's differential contributions relative to both prior works: (1) unified master equation $\mathbf{a}_{\text{out}} = W_{\text{eff}} \cdot \mathbf{a}_{\text{in}}$ covering feedforward NN, GNN, and Transformer under one principle; (2) multi-threshold field view with sequentially unlockable capabilities (vs. Lubana's single-threshold model); (3) $\sigma$ quantification via PR method with concrete falsifiable predictions; (4) explicit engineering framework positioning with a calibration roadmap.

## 10 Framework Positioning

NPM is an **engineering analogy framework**, not a proof system. Table 11 places it in the tradition of productive engineering frameworks that preceded rigorous theory.

Table 11: NPM in the context of established engineering frameworks

| **Framework** | **Basis** | **Params** | **Impact** | **NPM analog** |
|---|---|---|---|---|
| Darcy's Law (1856) | Empirical | $K$ (fitted) | 170 years | $\rho = N_B/\exp(\alpha\sigma)$ |
| Reynolds number (1883) | Dimensional | $\text{Re}_c$ (fitted) | Universal | $C_n = \rho/\rho_c$ |
| $k$-$\varepsilon$ model (1972) | Semi-empirical | 6 constants | CFD standard | $\alpha, \beta, \delta$ |

The evaluation criteria for NPM are those appropriate to engineering frameworks: logical consistency (verified), physical interpretability (demonstrated), parameter fittability (completed, Spearman= 0.70), and predictive generalization (partially demonstrated).

## 11 Boundaries of the PNM–NN Analogy

The value of an engineering analogy depends on clear recognition of its limits. This section explicitly delineates what can and cannot be mapped between PNM and NN.

## 11.1 Transferable Analogies

Table 12: PNM–NN transferable analogies

| **PNM phenomenon** | **NN counterpart** | **Basis** |
|---|---|---|
| Capillary entry pressure: $\Delta P < P_{c,\text{entry}} \Rightarrow Q = 0$ | Activation threshold: $x < 0 \Rightarrow \text{ReLU}(x) = 0$ | Both are hard thresholds; different pore sizes / neurons open at different values |
| Relative permeability: $k_{r,w} + k_{r,o} < 1$ (total flow < sum of single-phase flows) | Multilingual curse: per-language performance of multilingual model < monolingual | Conneau et al. [2020] empirically confirms; qualitatively matches relative permeability curves |
| Pressure-driven displacement: injected fluid applies pressure gradient, displaces gas phase, changes wetting interface | Fine-tuning: small directed dataset applies gradient perturbation, fills knowledge gaps, changes domain response | Both are small-slug / small-data driven local state redistribution |

## 11.2 Non-Transferable Differences

Table 13: PNM–NN non-transferable differences

| **PNM feature** | **NN difference** | **Implication** |
|---|---|---|
| Continuous viscous dissipation: $Q \propto \Delta P/\mu$ | Near-lossless information transfer; loss comes from threshold truncation, not continuous dissipation | No "information viscosity" exists in NNs; losses are discrete (ReLU cutoff, dropout), not continuous |
| Mass/volume conservation | Information can be copied and broadcast: one activation read by many downstream neurons without depletion | Attention reads all tokens simultaneously; fluid cannot be "read" in all pipes at once |
| Pore structure fixed (short-term) | Weights change during training: backpropagation continuously modifies "pore structure" | Training = fluid reshaping the rock as it flows; no direct PNM counterpart |
| Continuum assumption (Darcy's law requires REV) | Discrete structure: layers and neurons are discrete, violating the continuum assumption | Continuum exponents ($\beta$, etc.) borrowed from percolation may need NN-specific corrections |

## 11.3 Insights from the Differences

These non-transferable items are not weaknesses of the framework—they identify **unique features** of NNs that may seed new theory:

- **Information copyability** enables attention—it has no PNM counterpart and may require mathematics beyond classical transport theory.
- **Training reshapes structure**: NNs are "self-adaptive porous media" where flow sculpts the pore network. The closest geological process is dissolution, for which mature reactive-transport models exist.
- **Information interference** (contradictory data cancelling each other) maps most closely to reactive transport: $\partial c/\partial t = D\nabla^2 c - k \cdot c_A \cdot c_B$. This may provide a quantitative tool for understanding hallucination.

## 12 Future Directions

**B-path 1: Optimal $W_{\text{eff}}$ from maximum flow theory.** Given fixed parameter count, what $W_{\text{eff}}$ construction maximizes $C_{\text{eff}}$? If solved, the attention mechanism transitions from "engineering invention" to "physically necessary consequence of optimal information transport."

**B-path 2: Activation functions and capillary entry pressure.** ReLU thresholds are analogous to capillary entry pressure $P_{c,\text{entry}}$. The entry pressure distribution across pore throats is measurable via MICP curves; the optimal threshold distribution maximizing phase-transition sensitivity may be derivable from percolation critical theory.

**B-path 3: Relative permeability and multi-type information competition.** Capability loss in multilingual/multimodal training is analogous to $k_{r,1} + k_{r,2} < 1$ in multiphase flow. Establishing an NN version of the Corey/Brooks-Corey relative permeability model could quantitatively predict efficiency loss in mixed training.

**B-path 4: Three-phase transition at larger scales.** The pressurization–breakthrough–reorganization pattern (Section 8.3) has been observed on Pythia-70m only. Validating on Pythia-160m/410m/1B would test whether the phase transition step (as a fraction of total training) is scale-invariant—a prediction that percolation universality would suggest.

**B-path 5: Decoupling $P_{\min}$ from density.** The current formulation uses a single $\alpha$ in both $V(\sigma) = \exp(\alpha\sigma)$ and $P_{\min} = N_B \cdot \sigma^\alpha$, creating conflicting constraints (Section 7.3). Introducing an independent exponent for $P_{\min}$ or incorporating parameter count $P$ as an explicit variable in $C_{\text{eff}}$ are the two most promising resolution paths.

## 13 Conclusion

We have presented the Neural Percolation Model (NPM), a unified engineering framework mapping pore network physics onto neural network information propagation. A-layer checks confirm GCN/attention definitional consistency and reproduce classical percolation phase transitions. B-layer parameters have been fitted against 14 public models and Kaplan scaling data, achieving Spearman correlation 0.70 with MMLU and exact matching of the scaling exponent $N^{0.076}$. The fitted $\beta = 0.068$ departs significantly from the 3-D percolation value 0.41; we interpret this as reflecting the non-identity between $C_{\text{eff}}$ and test score, NN-specific topology, and the absence of parameter count $P$ in the current formulation.

MIS-based pore analysis of GPT-2 and Pythia-70m reveals that NN weight distributions share organizational shape with PNM pore-size distributions (rank correlation 0.98) despite belonging to different distribution families (Weibull $k = 1.1$ vs 2.5). Most significantly, Pythia-70m training exhibits a three-phase transition—pressurization, breakthrough, reorganization—with the coefficient of variation surging from 0.82 to 1.51 between steps 64k and 128k, directly analogous to giant connected component formation in percolation.

$C_n$ as a dimensionless criterion explains anomalies that parameter count alone cannot (e.g., BLOOM-176B underperforming LLaMA-7B), but objective measurement of $\sigma$ remains the bottleneck of the entire framework.

The value of NPM lies not in mathematical completeness but in providing practitioners a coherent physical vocabulary for reasoning about data, architecture, and capability jointly—a role analogous to that played by dimensional analysis and empirical correlations throughout the history of engineering.

**Code availability.** Implementation available at https://github.com/tiexinding/neural-percolation-model

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**Collaboration invitation.** This preprint is an open invitation for collaboration, not a claim of completeness. The author is actively seeking researchers with access to Pythia/OLMo checkpoints for quantitative validation, and theorists who can tighten the Transformer–PNM mapping. Contact: see GitHub repository above.

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